

Conditions on *wh-the-hell* licensing: Evidence from Colloquial Singapore English

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Abstract

The unacceptability of in-situ *wh-the-hell* has been observed since Pesetsky (1987). We investigate the acceptability of in-situ and partially-moved *wh-the-hell* in Colloquial Singapore English through an acceptability judgment experiment. The results reveal that in-situ *wh-the-hell* in single *wh*-questions is unacceptable, but partial *wh*-movement is able to host *the-hell*. We argue that this pattern follows from two independently supported components: 1. *wh-the-hell* needs to move to the Spec-CP so that the negative attitude bearer can be identified (see Chou (2012)); 2. Partial *wh*-movement in Singlish involves covert movement to Spec-CP while *wh*-in-situ is interpreted via unselective binding (Sato and Ngui, 2017).

Keywords: *wh-the-hell*, aggressively non-D-linked phrases, *wh*-movement, partial movement, experimental syntax

1 Introduction

Since Pesetsky (1987), one distributional restriction on aggressively non-D-linked phrases such as *the-hell* is their inability to stay in-situ. As the contrast in (1) shows, *the-hell* cannot modify an in-situ wh-phrase in English multiple questions (1a), but can modify the wh-phrase in matrix Spec-CP (1b).¹

- (1) a. * Who loves what the hell?
b. Who the hell loves what?

To account for the contrast, Den Dikken and Giannakidou (2002) argues that the higher wh-phrase *who* in (1a) intervenes in the licensing between the Q operator in CP and *wh-the-hell*, which leads to the unacceptability. An alternative analysis is one where *wh-the-hell* is required to occupy its scope-taking position in matrix Spec-CP in direct questions, which would also rule out the possibility of in-situ *wh-the-hell*. Chou (2012) argues for this requirement for the Mandarin counterpart of *the-hell*, *daodi*, which must move to Spec-CP to ascribe the negative attitude associated with *the-hell* to the speaker of the utterance in direct questions. Since both analyses can account for the contrast in (1), they cannot be teased apart in standard Englishes where wh-in-situ is only possible in multiple wh-questions. But they may be teased apart in varieties which allow wh-in-situ in single wh-questions as well as partial movement of wh-phrases to an intermediate Spec-CP, such as Colloquial Singapore English (Singlish).

¹We thank the editor and two anonymous reviewers for their helpful comments. We would also like to thank Michael Yoshitaka Erlewine, Nick Huang and audience members at the 4th International Conference on Theoretical East Asian Psycholinguistics, the 25th Seoul International Conference on Generative Grammar and Singapore Summer Meeting 2023 for their feedback on earlier versions of this project.

Singlish is a contact variety spoken in Singapore; it has a dominant English lexifier and has also been strongly influenced by its substrate languages such as Malay and southern Chinese languages (Sato, 2013). Singlish has three question-formation strategies, as shown in (2).² The *wh*-phrase either undergoes full *wh*-movement to matrix Spec-CP (2a), moves to an intermediate Spec-CP by partial movement (2b) or stays in-situ (2c). These three question types may be used interchangeably (Sato, 2013) (*pace* Lee, 2022).

- | | | | |
|-----|----|---|------------------|
| (2) | a. | What you think Natalie is baking at 3am ah? | full movement |
| | b. | You think what Natalie is baking at 3am ah? | partial movement |
| | c. | You think Natalie is baking what at 3am ah? | in-situ |
| | | ‘What do you think Natalie is baking at 3am?’ | |

Wh-the-hell which undergoes full movement is also attested in Singlish (3a) but the acceptability of *the-hell* in partial movement (3b) or *wh*-in-situ questions (3c) has not been investigated in detail. In addition, previous observations which identify this ban on in-situ *wh-the-hell* were based on multiple *wh*-questions in English rather than single *wh*-questions. Using an acceptability judgment task, we test whether partially-moved *wh*-phrases and *wh*-in-situ in single *wh*-questions are able to host *the-hell*.

- | | | |
|-----|----|--|
| (3) | a. | What the hell you think Natalie is baking at 3am ah? |
| | b. | You think what the hell Natalie is baking at 3am ah? |

²*Do*-support is optional in Singlish and *ah* is a question marker that blocks an echo question reading in *wh*-in-situ questions, i.e. ensures that *wh*-in-situ questions are interpreted as non-echo questions.

c. You think Natalie is baking what the hell at 3am ah?

‘What the hell do you think Natalie is baking at 3am?’

To preview the results, we find that in-situ *wh-the-hell* is unacceptable in single wh-questions in Singlish, but that partial movement is able to host *the-hell*. We propose that this pattern can be accounted for by combining Chou (2012)’s analysis of *the-hell* and the independently proposed syntax of the three question-formation strategies in Singlish. Specifically, we argue that *the-hell* must move to matrix Spec-CP to ascribe its negative attitude to the speaker and the movement of *the-hell* is parasitic on the (overt/covert) movement of the wh-phrase. When the wh-element moves to matrix Spec-CP, the *the-hell*, moving along with it, is licensed. In full wh-movement questions, *wh-the-hell* moves to Spec-CP overtly. In partial movement questions, *wh-the-hell* moves to the intermediate Spec-CP overtly and then to the matrix Spec-CP covertly, as argued by Sato and Ngui (2017). On the other hand, Singlish wh-in-situ is interpreted via unselective binding. Since the *wh-the-hell* does not move to Spec-CP, the negative attitude ascription does not take place and *the-hell* is not licensed.

This paper is structured as follows. We present the acceptability judgment experiment in §2. In §3, we present our analysis and derivations, and also show that two alternative accounts do not capture the empirical facts in Singlish.

2 Experimental investigation on *wh-the-hell* in Singlish

2.1 Design, materials, and procedure

We aim to investigate whether the ban on in-situ *wh-the-hell* is also observed for single *wh*-questions in Singlish and if partially-moved *wh*-phrases are able to host *the-hell* through an acceptability judgment experiment. *Wh*-in-situ and partial movement are tested in separate comparisons, and each comparison uses a 2×2 factorial design (see also Sprouse, Wagers, and Phillips (2012); Sprouse, Caponigro, et al. (2016) which investigates island effects using a similar methodology). In the *wh*-in-situ comparison, the factors *WhType* (*Wh*, *WhHell*) and *Strategy* (*Long*, *Situ*) are crossed to create the four conditions in (4).³ Since *wh-the-hell* in questions with full *wh*-movement such as (4b) is attested in Singlish, *Long* conditions which contain full movement are used as a grammatical baseline to assess if there is a penalty when *the-hell* modifies the in-situ *wh*-phrase in (4d).

- | | | | |
|-----|----|--|-------------|
| (4) | a. | What you think Natalie is baking at 3am ah? | Wh-Long |
| | b. | What the hell you think Natalie is baking at 3am ah? | WhHell-Long |
| | c. | You think Natalie is baking what at 3am ah? | Wh-Situ |
| | d. | You think Natalie is baking what the hell at 3am ah? | WhHell-Situ |
| | | ‘What (the hell) do you think Natalie is baking at 3am?’ | |

The 2×2 design controls for other factors which may have an independent effect on sentence acceptability. Taking the *wh*-in-situ comparison as an example, we plot

³The conditions contain embedded clauses to ensure that these materials are parallel to the partial movement conditions in (6), as both partial *wh*-movement and *wh*-in-situ were tested in the same experiment.

hypothetical condition means for (4) on the interaction plots in figure 1. If there is a penalty associated with in-situ *wh-the-hell*, an additional decrease in acceptability should be observed for the WhHell-Situ condition (4d) after costs associated with the presence of *the-hell* (WhType) and question-formation type (Strategy) are controlled for. This is reflected by the superadditive effect illustrated in figure 1a. Statistically, we would then expect to observe a significant interaction of WhType and Strategy. Conversely, if the costs associated with the presence of *the-hell* (WhType) and question-formation type (Strategy) are linearly additive and can account for the decrease in acceptability observed for (4d), we would conclude that in-situ *wh-the-hell* does not incur an additional penalty. We would then expect to see an interaction plot like figure 1b, and the statistical analysis should not find an interaction of WhType and Strategy.

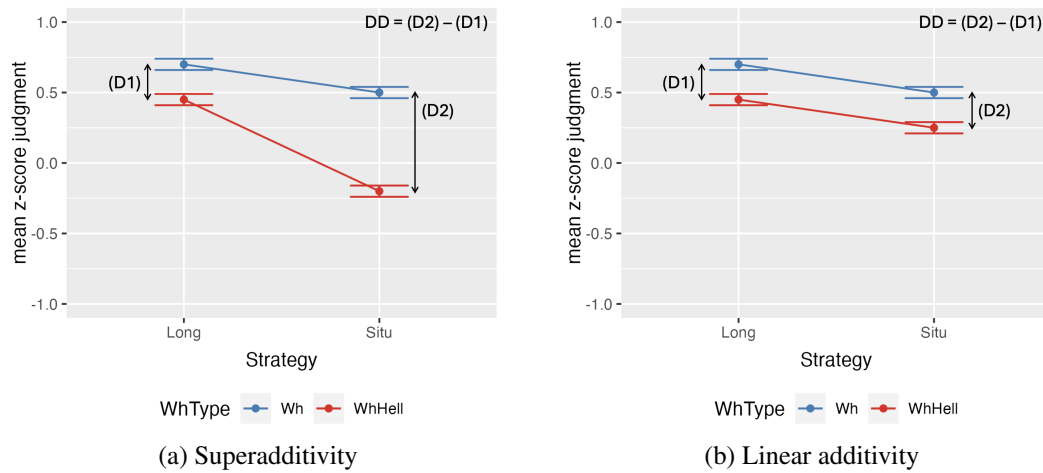


Figure 1: Predicted interaction plots for wh-in-situ comparison

In addition, the size of the penalty may be quantified numerically using a *differences-in-differences* (DD) score (Maxwell and Delaney, 2003). A DD score calculation using the hypothetical condition means in figure 1a is shown in (5). As

shown on the interaction plot in figure 1a, (D1) reflects the difference between Wh-Long and WhHell-Long conditions (5a), that is, the penalty of adding *the-hell* to full wh-movement questions. (D2) represents the difference between Wh-Situ and WhHell-Situ conditions (5b), which is the penalty of adding *the-hell* to wh-in-situ questions. The difference between these two difference scores is then calculated, as shown in (5c). The DD score is positive if *the-hell* induces a larger penalty in wh-in-situ questions relative to full wh-movement questions, and this would also be indicative of a superadditive interaction. If there is no interaction of WhType and Strategy, we expect a DD score of zero or close to zero.

(5) Calculating the DD score in figure 1a:

a. $(D1) = \text{Wh-Long} - \text{WhHell-Long}$

Wh-Long: What you think Natalie is baking at 3am ah?

WhHell-Long: What the hell you think Natalie is baking at 3am ah?

b. $(D2) = \text{Wh-Situ} - \text{WhHell-Situ}$

Wh-Situ: You think Natalie is baking what at 3am ah?

WhHell-Situ: You think Natalie is baking what the hell at 3am ah?

c. $DD = (D2) - (D1) > 0$

As for the partial movement comparison, the factors WhType (Wh, WhHell) and Strategy (Long, Partial) are crossed to create the conditions in (6). Like the wh-in-situ comparison, Long conditions are used as a grammatical baseline as *wh-the-hell* in full movement questions is independently observed in Singlish. If *the-hell* induces a penalty in partial wh-movement questions like (6d), we expect to find a significant

interaction of WhType and Strategy, a superadditive effect when the condition means are plotted on an interaction plot, like figure 1a, as well as a positive DD score. If no such penalty is observed, we would expect an additive effect like figure 1b with a DD score close to zero, and no interaction between WhType and Strategy.

- | | | | |
|-----|----|--|----------------|
| (6) | a. | What you think Natalie is baking at 3am ah? | Wh-Long |
| | b. | What the hell you think Natalie is baking at 3am ah? | WhHell-Long |
| | c. | You think what Natalie is baking at 3am ah? | Wh-Partial |
| | d. | You think what the hell Natalie is baking at 3am ah? | WhHell-Partial |
- ‘What (the hell) do you think Natalie is baking at 3am?’

Both wh-in-situ and partial movement comparisons are tested in the same experiment. As both comparisons are made relative to the same Long conditions, the experiment contains 6 test conditions. Each participant saw 2 items per condition, which results in 12 test items in total. 24 filler items were included so that items across the whole experiment are balanced for acceptability, and 3 check items were added to exclude participants who are not attentive to the task or are not native speakers.⁴ The experiment was hosted on PCIBex (Zehr and Schwarz, 2018). Items were presented to participants one sentence at a time, and participants were required to rate the naturalness of each item as Singlish using a 100-point slider.

⁴The three check items comprised 2 right dislocation items (i,ii) and one attention check item. Particles such as *sia* are needed to license right dislocation in Singlish (Chan, 2023), and 32 out of 34 participants in this study (94%) observed this contrast. The two participants who gave (ii) a higher rating than (i) were excluded from the analysis. The attention check item instructed participants to click on the left end of the slider and participants who did not click as instructed were excluded from the analysis. No participants were excluded based on their responses to the attention check item.

- (i) Capable *sia* Yi Lin’s marketing intern.
‘Yi Lin’s marketing intern is capable.’ (Particle *sia* expresses the speaker’s surprise)
- (ii) * Capable Yi Lin’s marketing intern.
Intended: ‘Yi Lin’s marketing intern is capable.’

35 Singlish native speakers were recruited through the National University of Singapore's research participation platform. Participants grew up and attended primary and secondary school in Singapore and also self-reported that they use Singlish on a daily basis. Each participant was paid SGD5 after completing the experiment. One participant was excluded as they completed the online survey twice, and two other participants were excluded as they did not show the expected contrast in the check items. Hence, we analyzed the responses from 32 participants.

Wh-in-situ (4) and partial movement (6) were analysed in separate comparisons. Participants' raw ratings were z-score transformed and linear mixed effects models were constructed for each comparison in RStudio (RStudio Team, 2024). Treatment contrast coding was applied to the two factors WhType (Wh: 0, WhHell: 1) and Strategy (Wh-in-situ comparison: Long: 0, Situ: 1; Partial movement comparison: Long: 0, Partial: 1). In each model, WhType, Strategy and the interaction term were included as fixed effects with by-participant and by-item random intercepts and slopes. Random terms were then removed from the model with maximal random effects until the model converged. The exact model used is specified below the table that presents the model results for each comparison.⁵

2.2 Results

We first present the results for the wh-in-situ comparison. The mean raw ratings (out of 100) and z-score transformed ratings for each condition (as well as their respective standard deviations) are presented in table 1, and the z-score transformed ratings are

⁵The materials, data, and analysis of the experiment can be found at <https://osf.io/juf9s/overview>

plotted in figure 2. The error bars represent ± 1 standard errors. A superadditive effect is observed in the interaction plot, and the WhHell-Situ condition has the lowest mean rating relative to the other three conditions. This results in a large positive DD score of 1.15.

Condition	WhType	Strategy	Mean raw	SD raw	Mean z-score	SD z-score
WhHell-Long	WhHell	Long	78.94	25.22	0.39	0.62
Wh-Long	Wh	Long	83.61	22.95	0.56	0.52
WhHell-Situ	WhHell	Situ	37.78	29.67	-0.78	0.72
Wh-Situ	Wh	Situ	83.17	19.11	0.54	0.53

Table 1: Condition means and standard deviations (wh-in-situ)

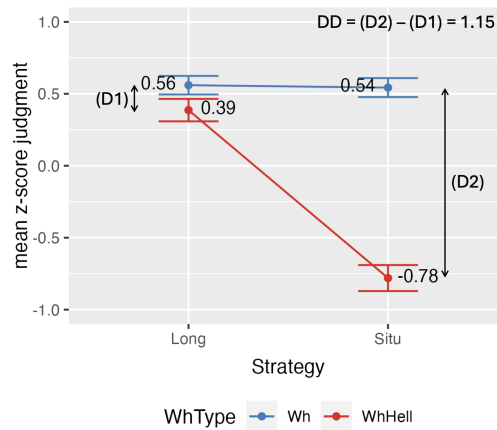


Figure 2: Interaction plot for wh-in-situ analysis

Table 2 summarizes the model results. There is a significant interaction of WhType and Strategy ($p < 0.001$). There is no main effect of WhType or Strategy (WhType: $p = 0.122$; Strategy: $p = 0.807$). Taken together, the results are indicative of a penalty associated with in-situ *wh-the-hell*. Even in single wh-in-situ questions, *the-hell* cannot modify the in-situ wh-element.

	Estimate	Std. Error	df	t value	Pr(> t)
Intercept	0.571	0.093	12.809	6.086	< 0.001
WhType	-0.191	0.116	14.260	-1.646	0.122
Strategy	-0.030	0.122	21.880	-0.247	0.807
WhType:Strategy	-1.161	0.231	16.036	-5.018	< 0.001

Table 2: Model results for wh-in-situ vs. full wh-movement
model: $z\text{scores} \sim \text{WhType} * \text{Strategy} + (1 + \text{WhType} * \text{Strategy} \mid \text{subject}) + (1 + \text{WhType} * \text{Strategy} \mid \text{lexical})$

We now look at the results for the partial movement comparison. The mean raw ratings and z-score transformed ratings (and their standard deviations) are presented in table 3 and the z-score transformed ratings are plotted on the interaction plot in figure 3, with error bars for ± 1 standard errors.

Although the WhHell-Partial condition has the lowest mean rating among the four conditions in this comparison, the costs which are independently associated with the presence of *the-hell* as well as partial movement appear linearly additive in deriving the slight penalty observed for the WhHell-Partial condition. This is supported by the parallel lines on the interaction plot in figure 3, and the DD score (-0.02) which is close to zero.

Condition	WhType	Strategy	Mean raw	SD raw	Mean z-score	SD z-score
WhHell-Long	WhHell	Long	78.94	25.22	0.39	0.62
Wh-Long	Wh	Long	83.61	22.95	0.56	0.52
WhHell-Partial	WhHell	Partial	58.05	30.00	-0.19	0.73
Wh-Partial	Wh	Partial	63.58	30.26	-0.03	0.73

Table 3: Condition means and standard deviations (partial movement)

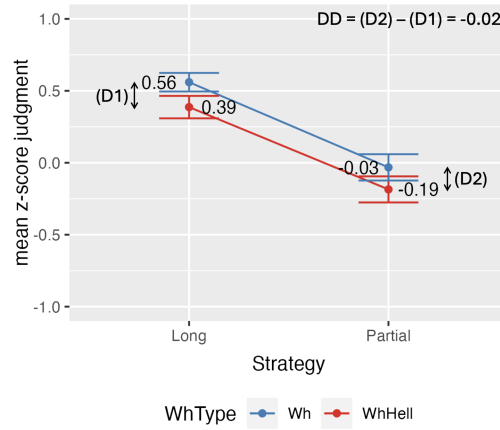


Figure 3: Interaction plot for partial movement analysis

Table 4 summarizes the model results. Strategy is a main effect ($t = -0.594, p < 0.001$): Long conditions are generally rated more favourably than Partial conditions. There is no main effect of WhType ($p = 0.09$). Crucially, the interaction of WhType and Strategy does not reach significance ($p = 0.88$). As such, there does not appear to be an additional penalty incurred when partial wh-movement hosts *the-hell*.⁶ Note that although Wh-Partial conditions are rated around the middle of the scale, they have a higher mean rating than our ungrammatical fillers (mean raw score: 43.5; mean z-score: -0.61).

	Estimate	Std. Error	df	t value	Pr(> t)
Intercept	0.560	0.088	55.161	6.375	< 0.001
WhType	-0.172	0.100	185.968	-1.727	0.086
Strategy	-0.594	0.115	73.737	-5.153	< 0.001
WhType:Strategy	0.021	0.141	187.439	0.149	0.882

Table 4: Model results for partial wh-movement vs. full wh-movement
model: $zscores \sim WhType * Strategy + (1 + Strategy | subject) + (1 | lexical)$

⁶Sato (2013) speculates that partial movement of the bare wh-phrase is marginal as wh-movement and wh-in-situ are more robust question-formation strategies.

3 Accounting for the Singlish contrast

The results of the experiment establish the empirical generalization that in-situ *wh-the-hell* is unacceptable for single wh-questions in Singlish, and to the extent that partial wh-movement is accepted, it is able to host *the-hell*.⁷ Our analysis for the possibility of *wh-the-hell* in partial/full wh-movement questions but not in wh-in-situ questions combines two independently supported components: 1. a licensing condition on *the-hell* which requires its negative attitude to be ascribed to the speaker of the utterance in direct questions, and 2. the syntax of the three wh-question strategies in Singlish. We elaborate more on the syntax of Singlish wh-questions in §3.1 and the negative attitude of *wh-the-hell* in §3.2, and sketch out the derivations for Singlish *wh-the-hell* in §3.3.

3.1 Syntax of wh-questions in Singlish

Three types of wh-questions are attested in Singlish, as shown in (2) repeated below, and they may be used interchangeably (Sato, 2013). We follow Sato and Ngui (2017)’s analysis of the three question-formation strategies.

- | | | |
|-----|--|---------------------|
| (2) | a. What you think Natalie is baking at 3am ah? | full wh-movement |
| | b. You think what Natalie is baking at 3am ah? | partial wh-movement |
| | c. You think Natalie is baking what at 3am ah? | wh-in-situ |

⁷We note that Jackendoff and Audring (2020) mention other English intensifiers such as {the heck/the fuck/the devil/the dickens/in the world/in God’s name/etc.} (See also Hoeksema and Napoli (2008)). Not all of these expressions are attested in Singapore English. For instance, we only find instances of ‘what the hell/the heck/the fuck/in the world’ but not other intensifiers in The Corpus of Singapore English Messages (Gonzales et al., 2023). To the extent that these intensifiers are observed in Singlish and express the speaker’s negative attitude, we predict that they will behave like Singlish *wh-the-hell*. We thank the editor for raising this point.

‘What do you think Natalie is baking at 3am?’

Full wh-movement questions in Singlish have the same derivation as those in its lexifier, English, except that *do*-support is optional in Singlish and is omitted in (2a). As shown in (7), the derivation for (2a), matrix C has a strong [+wh] feature which requires checking by a wh-phrase in its specifier. The wh-phrase undergoes successive cyclic movement to matrix Spec-CP (7).

(7) Derivation for full wh-movement:

Successive cyclic movement to matrix Spec-CP:

[_{CP} what_i_[+wh] C_[+Q, +wh] you think [_{CP} t_i Natalie is baking t_i at 3am ah]]

Analyses for wh-in-situ questions may take an unselective binding or covert movement approach. Under an unselective binding account, the wh-phrase contains a nominal variable which allows it to be bound by a Q operator in the CP domain to obtain its interrogative interpretation (Pesetsky, 1987; Tsai, 1994). The covert movement analysis assumes that wh-in-situ questions share a similar derivation as their wh-movement counterparts, but movement takes place at LF and is thus covert (Huang, 1982). For reasons to be discussed below, we adopt an unselective binding analysis for Singlish wh-in-situ, rather than a covert movement analysis, following Sato and Ngui (2017). The Q operator binds the wh-phrase in (2c), as shown in (8).

(8) Derivation for wh-in-situ:

Unselective binding of wh-phrase:

[_{CP} Q You think [_{CP} Natalie is baking what at 3am ah]]

Sato and Ngui (2017) point to the availability of non-interrogative readings of wh-words in (9a-c) as evidence for unselective binding. Sentences in (9a-c) are declarative sentences and the non-interrogative use of *what*, *who* and *where* suggests that these wh-words are variables which are bound by a non-interrogative operator. In contrast, *why* is not a wh-variable and cannot be bound, and a non-interrogative reading of *why* is not possible in (9d). This also explains the unavailability of in-situ *why* questions in (10b), since *why* is not a wh-variable and unselective binding cannot

apply to it. Both contrasts in (9) and (10) cannot be straightforwardly explained by a covert wh-movement analysis.

- (9) a. I what also never eat. Sato and Ngui (2017), ex.18

‘For all x, I didn’t eat x.’

- b. I who also never meet.

‘For all x, I didn’t meet x.’

- c. I where also never go.

‘For all x, I didn’t go to x.’

- d. * I why also never cheat.

‘For all x, I didn’t cheat because of x.’

- (10) a. Why Mary must leave early? (Chow, 1995)

- b. * Mary must leave early why?

The contrast in (11) is another piece of evidence for unselective binding. Although island effects are observed in (11a) when the wh-phrase undergoes overt movement across the Complex NP island boundary, as in (12), the question is no longer unacceptable when the wh-phrase is in-situ (11b). This is expected if unselective binding applies (13) and no movement across the island boundary takes place.

- (11) a. * What John like the man that think Mary eat? (Adapted from Sato and Ngui, 2017)

Intended: ‘What is the thing x such that John likes the man that thinks Mary eats x?’

b. John like the man that think Mary eat what?

‘What is the thing x such that John likes the man that thinks Mary eats
x?’

(12) Derivation for (11a):

*What_i John like [_{DP} the man that think t_i Mary eat t_i]

 A solid line with upward-pointing arrows at both ends connects the trace t_i in the embedded clause to the wh-phrase What_i in the matrix clause.

(13) Derivation for (11b):

[_{CP} Q John like [_{DP} the man that think Mary eat what]]

 A dashed line with downward-pointing arrows at both ends connects the wh-phrase what in the embedded clause to the operator Q in the matrix Spec-CP position.

We now come to partial movement questions which involve overt movement of the wh-phrase to an intermediate Spec-CP. The derivation for (2b) is presented in (14). Matrix C is specified with [+wh] and the wh-phrase undergoes an overt movement step to intermediate Spec-CP on its way to check the [+wh] feature on matrix C (14a). This representation then undergoes Spell-out. The wh-phrase cannot stop at this intermediate Spec-CP position as unselective binding does not target non-argument positions. Thus, the wh-phrase must then undergo a covert movement step to matrix Spec-CP (14b).

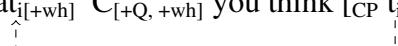
(14) Derivation for partial wh-movement:

a. Overt movement step to intermediate Spec-CP:

[_{CP} C_[+Q, +wh] you think [_{CP} what_i[_{+wh}] Natalie is baking t_i at 3am ah]]

 A solid line with upward-pointing arrows at both ends connects the trace t_i in the embedded clause to the wh-phrase what_i[_{+wh}] in the intermediate Spec-CP position.

b. Covert movement step to matrix Spec-CP:

[_{CP} what_i[_{+wh}] C_[+Q, +wh] you think [_{CP} t_i Natalie is baking t_i at 3am ah]]

 A dashed line with upward-pointing arrows at both ends connects the trace t_i in the embedded clause to the wh-phrase what_i[_{+wh}] in the matrix Spec-CP position.

Sato and Ngui (2017) claims that the island effect observed in (15) provides evidence for this covert movement step. In (15), the *wh*-element is moved to the embedded Spec-CP. As the embedded CP is contained in a complex NP, there is an island boundary between the matrix Spec-CP and the *wh*-element. As shown in (11b), unselective binding is not sensitive to islands, hence the unacceptability of (15) suggests that the partially moved *wh*-element is not bound by the Q operator but has covertly moved to matrix Spec-CP. The unacceptability thus results from the covert movement crossing an island boundary. This is illustrated in the derivation in (16).

- (15) * John like the man that think what Mary eat? (Adapted from Sato and Ngui, 2017)

Intended: ‘What is the thing *x* such that John likes the man that thinks Mary eats *x*?’

- (16) Predicted derivation for (15):

- a. Overt movement step to intermediate Spec-CP:

[_{CP} John like [_{DP} the man that think what_{*i*} Mary eat t_{*i*}]]



A solid line with an upward arrow at the end, connecting the trace *t_i* to the *wh*-element *what_i*.

- b. Covert island-violating movement step to matrix Spec-CP:

[_{CP} what_{*i*} John like [_{DP} the man that think t_{*i*} Mary eat t_{*i*}]]



A dashed line with an upward arrow at the end, connecting the trace *t_i* to the *wh*-element *what_i*.

Cole and Hermon (1998) also observe a similar pattern in Malay, one of the substrate languages of Singlish which has influenced Singlish partial movement (Sato, 2013). Like Singlish, island effects are not observed when the *wh*-phrase is in-situ in the Complex NP in Malay (17a), but partial movement of the *wh*-phrase is not possible even when this movement step does not cross the island boundary (17b).

This is compatible with an analysis in which *wh*-in-situ is interpreted by unselective binding but the derivation for partial movement questions involves a covert movement step.

- (17) a. Kamu sayang [perempuan yang Ali fikir [yang telah makan apa]]
 You love woman that Ali thinks that already eat what
 ‘You love the woman who Ali thinks ate what?’
- b. * Kamu sayang [perempuan yang Ali fikir [apa_i yang telah makan t_i]]
 You love woman that Ali thinks what that already eat
 ‘You love the woman who Ali thinks ate what?’

To sum up, the *wh*-element in Singlish is interpreted at Spec-CP in both full *wh*-movement and partial *wh*-movement questions, while the in-situ *wh*-element is interpreted at its base-generated position.

3.2 *Negative attitude of the-hell*

Previous work focusing on the semantic and pragmatic properties of *the-hell* finds that *wh-the-hell* questions are generally associated with a negative attitude or negative bias (See Den Dikken and Giannakidou (2002)). *Wh-the-hell* conveys the speaker’s surprise (Pesetsky, 1987), ignorance concerning the answer (Martin, 2020) and belief that any possible answer is improbable (Rawlins, 2008). Although *the-hell* may be used in rhetorical questions (See Ippolito (2024)), the negative attitude of *the-hell* is still compatible with information-seeking questions. For instance, the sentence in (18) is not rhetorical and expects an answer. In a recent analysis by Ippolito (2024), *the-hell* carries a conventional implicature such that in the case of (18), *the-hell* contributes the meaning that for every person *x*, the speaker has reason to doubt that *x* bought that

book.

- (18) Who the hell bought that book?

The negative attitude associated with *the-hell* is also observed in Singlish, and in direct information-seeking questions such as (3a) repeated here, the speaker of the utterance bears the negative attitude.⁸

- (3a) What the hell you think Natalie is baking at 3am ah?
'What the hell do you think Natalie is baking at 3am?'

Following Chou (2012), we incorporate the negative attitude of *the-hell* into the syntactic computation, to explain the contrast between partially-moved and in-situ *wh-the-hell*. Inspired by Huang and Ochi (2004), Chou (2012) proposes a system of Point-of-View (POV) features and POV operators for the Mandarin equivalent of *the-hell*, *daodi*. Building on Chou (2012), we assume that *the-hell* in standard English and Singlish contains an unvalued POV feature [*ud*] which is checked and valued [+d] in a Spec-head relationship with the POV operator in matrix CP.⁹

⁸Huang and Ochi (2004) claims that the negative attitude holder may either be the external speaker (speaker of the utterance) or the internal speaker. The internal speaker refers to the matrix subject in embedded *wh-the-hell* questions such as (i). In our experiment, the negative attitude holder is always the external speaker who produces the utterance, which we hereafter refer to as 'speaker' for simplicity.

(i) John wonders why the hell Bill said that. (Huang and Ochi, 2004)

⁹Our implementation differs from Chou (2012), who claims that the unvalued POV feature [*ud*] may be valued [+d] or [-d] by Agree (via the probe-goal relationship between the POV operator and *daodi* 'the hell') to capture the observation that *daodi* may only be embedded under second-person (but not third-person) matrix subjects.

(i) {Ni/*Zhangsan} renwei [Lisi daodi xihuan shenme]?
you/Zhangsan think Lisi THE-HELL like what
'What the hell do you/Zhangsan think Lisi likes?' (Adapted from Chou, 2012)

This sensitivity to the matrix subject's person features is not observed in Singlish; *wh-the-hell* is still acceptable when it is extracted from the embedded clause across a third-person matrix subject (ii). Hence, we do not use two values for the POV feature in our analysis and we also assume that the POV feature is checked in a Spec-head relationship with matrix C. That said, our proposal can still be made compatible

Crucially, *wh-the-hell* must move to matrix Spec-CP to check the POV feature, and this ascribes the negative attitude of *the-hell* to the speaker (Chou, 2012). We follow Merchant (2002) in assuming that *the-hell* is a modifier which adjoins to the wh-head. As such, *the-hell* cannot undergo movement on its own and its movement must be parasitic on the movement of the wh-phrase. This assumption departs from Huang and Ochi (2004) and Chou (2012) who assume that *daodi* ‘the hell’ in Mandarin Chinese is freely able to undergo covert movement. We suggest that the difference between English/Singlish *the-hell* and Mandarin *daodi* arises because *the-hell* must adjoin to the wh-phrase in English and Singlish, while *daodi* in Mandarin need not, as (19) shows.

- (19) Ni renwei [Lisi daodi xihuan shenme]?
you think Lisi THE-HELL like what
‘What the hell do you think Lisi likes?’ (Chou, 2012)

3.3 Derivations for Singlish wh-the-hell

With the two components in place, we can account for the distribution of *wh-the-hell* in Singlish (and English). We assume, following Chou (2012), that *the-hell* must move to matrix Spec-CP by the end of the derivation to check its POV feature and ascribe its negative attitude to the speaker in direct questions. Since *the-hell* is parasitic on the overt/covert movement of the wh-phrase to matrix Spec-CP, full and partial wh-movement questions are able to license *the-hell*, as *the-hell* is in Spec-CP by the end of the derivation. In wh-in-situ questions involving unselective binding, on the other hand, the wh-phrase does not undergo overt or covert movement to Spec-CP. As

with a probe-goal implementation of Agree.

- (ii) What the hell Brandon think Natalie is baking ah?
‘What the hell does Brandon think Natalie is baking?’

a result, *the-hell* is not in Spec-CP and the negative attitude bearer cannot be identified.

We now sketch out derivations for Singlish *wh-the-hell* in full wh-movement, wh-in-situ and partial wh-movement questions. The derivation for *wh-the-hell* questions with full wh-movement (3a) repeated below is schematised in (20). *The-hell* adjoins to the wh-phrase in its base-generated position and *wh-the-hell* undergoes a movement step to embedded Spec-CP (20a). The POV feature on *the-hell* is still unvalued at this point in the derivation. Next, the POV operator is merged in the matrix clause (20b), and *wh-the-hell* moves to matrix Spec-CP to check its POV feature (20c). The negative attitude of *wh-the-hell* can be ascribed to the speaker as *the-hell* reaches its scope-taking position by overt movement of the wh-phrase.

- (3a) What the hell you think Natalie is baking at 3am ah?
‘What the hell do you think Natalie is baking at 3am?’

(20) Derivation for *wh-the-hell* with full wh-movement:

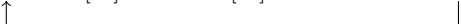
- a. Movement of *wh-the-hell* to embedded Spec-CP:

[_{CP} [what the hell]_{i[ud]} Natalie is baking t_i at 3am ah]


- b. POV operator is merged:

[_{CP} POV-op_[+d] you think [_{CP} [what the hell]_{i[ud]} Natalie is baking t_i at 3am ah]]

- c. Movement of *wh-the-hell* to matrix Spec-CP and feature-checking:

[_{CP} [what the hell]_{i[+d]} POV-op_[+d] you think [_{CP} t_i Natalie is baking t_i at 3am ah]]


As Singlish wh-in-situ is interpreted by unselective binding, (3c) would have the

derivation in (21). After *the-hell* adjoins to the wh-phrase in its base-generated position (21a), the Q operator binds the wh-phrase in its base-generated position (21b). As such, the wh-phrase does not undergo covert movement, and it is not possible for *the-hell*, which is adjoined to the wh-phrase, to reach the matrix POV operator to check its POV feature. The negative attitude of *the-hell* cannot be ascribed to the speaker and the resulting representation cannot be interpreted.

(3c) * You think Natalie is baking what the hell at 3am ah?

Intended: ‘What the hell do you think Natalie is baking at 3am?’

(21) Derivation for in-situ *wh-the-hell* (object):

a. *the-hell* is base-generated with the wh-phrase:

*_{[CP POV-op_[+d] Q} You think Natalie is baking [what the hell]_[ud] at 3am ah]

b. Unselective binding:

*_{[CP POV-op_[+d] Q} You think Natalie is baking [what the hell]_[ud] at 3am
ah]

We observe the same restriction on in-situ *wh-the-hell* in subject wh-questions, as (22) shows.¹⁰ The wh-phrase is in-situ in IP as it follows the overt complementizer *that*.¹¹ Although subject wh-in-situ questions such as (22a) are attested in Singlish, the question is no longer acceptable when *the-hell* modifies the wh-phrase in (22b).¹²

(22) a. You that time heard that who went hospital for surgery ah?

‘Who is the person *x* such that you heard that *x* went to the hospital for surgery?’

b. * You that time heard that who the hell went hospital for surgery ah?

Intended: ‘Who the hell is the person *x* such that you heard that *x* went to the hospital for surgery?’

Our proposal is also able to explain this contrast. As the in-situ wh-phrase is interpreted by unselective binding, the wh-phrase does not undergo covert movement.

This is illustrated in the derivation in (23) for (22b). As Q binds the wh-phrase, the

¹⁰Singlish allows for high temporal modifiers such as *that time* (which indicates some time in the past) that intervene between subject and predicate.

¹¹We do not consider matrix subject wh-questions such as (i) which are acceptable as we cannot ensure that the wh-phrase is in-situ since matrix clauses do not contain an overt complementiser in Singlish.

(i) Who the hell went hospital for surgery ah?

¹²As the experiment in §2 only tested object wh-questions, informal acceptability judgments for (22) were collected from 7 native Singlish speakers.

wh-phrase does not move and *the-hell* cannot move to its feature-checking position in matrix Spec-CP (23b). As a result, the negative attitude cannot be identified and (22b) is unacceptable.¹³

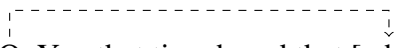
(23) Derivation for in-situ *wh-the-hell* (subject):

a. *the-hell* is base-generated with the wh-phrase:

*<sub>[CP POV-op_[+d] Q You that time heard that [who the hell]_[ud] went hospital
for surgery ah]</sub>

b. Unselective binding:

*<sub>[CP POV-op_[+d] Q You that time heard that [who the hell]_[ud] went hospital
for surgery ah]</sub>



Now we turn to patterns in partial wh-movement questions, such as (3b), repeated below. *The-hell* adjoins to the wh-phrase and its POV feature is originally unvalued, and *wh-the-hell* moves to intermediate Spec-CP (24a). After the matrix clause and POV operator are merged, this representation undergoes Spell-out (24b). *Wh-the-hell* then undergoes covert movement to matrix Spec-CP in (24c). The POV feature on *the-hell* is checked and its negative attitude is ascribed to the speaker. The resulting representation can be interpreted as the negative attitude bearer is correctly identified.

(3b) You think what the hell Natalie is baking at 3am ah?

‘What the hell do you think Natalie is baking at 3am?’

(24) Derivation for partially-moved *the-hell*:

a. Overt movement step to intermediate Spec-CP:

¹³We thank Michael Erlewine (p.c.) for the suggestion to look into subject wh-in-situ questions.

[_{CP} [what the hell]_{i[ud]} Natalie is baking t_i at 3am ah]


- b. POV operator is merged:

[_{CP} POV-op_[+d] you think [_{CP} [what the hell]_{i[ud]} Natalie is baking t_i at 3am ah]]

- c. Covert movement step to matrix Spec-CP and feature-checking:

[_{CP} [what the hell]_{i[+d]} POV-op_[+d] you think [_{CP} t_i Natalie is baking t_i at 3am ah]]


This proposal comes together because of two independent properties of *wh-the-hell* questions in Singlish. First, the negative attitude associated with *the-hell* must be ascribed to the speaker in direct questions by movement of *the-hell* to matrix Spec-CP to check its POV feature. This requirement on *the-hell* is independently motivated by Chou (2012). Second, due to differences in the syntax of full wh-movement, partial wh-movement and wh-in-situ, only full and partial wh-movement questions contain a movement step to matrix Spec-CP in their derivation, while there is no movement to matrix Spec-CP in wh-in-situ questions as unselective binding applies (Sato, 2013). By combining these two properties of *wh-the-hell* questions, we are able to account for the distribution of *wh-the-hell* in Singlish as well as English.

3.4 Alternative analyses

In this section, we review two alternative accounts for the ban on in-situ *wh-the-hell* in the literature: Den Dikken and Giannakidou (2002)'s intervention effects account and Vu and Lohiniva (2020)'s AttP account, and show that these analyses are insufficient to account for the Singlish data.

3.4.1 Den Dikken and Giannakidou (2002)

Den Dikken and Giannakidou (2002) claim that *wh-the-hell* is a polarity item (PI) as it is similar in distribution to other PIs such as *any*. Under this account, the ungrammaticality of in-situ *wh-the-hell* in (26a) is attributed to intervention effects. Following the assumption that PIs need to be in the immediate scope of their licenser (Linebarger, 1987), *who the hell* in (25a) is licensed as it is in the immediate scope of the Q operator, as shown in (25b). On the other hand, *who the hell* in (26a) is not licensed because the subject wh-phrase *who* intervenes and blocks the licensing relation between Q and *wh-the-hell*, as schematized in (26b).

(25) a. Who the hell is in love with who?

b. [Q ... [**who the hell** ... [who]]]

(26) a. * Who is in love with who the hell? Den Dikken and Giannakidou (2002),

ex.71a

b. * [Q ... [**who** ... [who the hell]]]

ex.73b

This account makes the correct predictions for in-situ *wh-the-hell* in English multiple wh-questions, as the higher wh-phrase will always act as an intervenor between Q and the lower wh-phrase. However, this analysis is unable to explain the ban on in-situ *wh-the-hell* in single wh-questions in Singlish. As (3c) shows, it is not possible for *the-hell* to modify wh-in-situ in single wh-questions, although there is no wh-phrase that intervenes between Q and *wh-the-hell* in (27).

(3c) * You think Natalie is baking what the hell at 3am ah?

Intended: 'What the hell do you think Natalie is baking at 3am?'

(27) * [Q ... [Natalie ... what the hell]]

This is also observed for subject *wh*-in-situ questions such as (22b). Even though Q should be able to license *the-hell* in (28), this is not observed empirically as *the-hell* cannot modify an in-situ subject *wh*-phrase.¹⁴

(22b) * You that time heard that who the hell went hospital for surgery ah?

‘Who the hell is the person *x* such that you heard that *x* went to the hospital for surgery?’

(28) * [Q You ... [who the hell ...]]

3.4.2 Vu and Lohiniva (2020)

Vu and Lohiniva (2020) build on Huang and Ochi (2004)’s proposal that *wh-the-hell* questions contain an Attitude Phrase (AttP) within IP. AttP has a [+wh] feature which must be checked by a *wh*-phrase in its specifier. Crucially, they argue that *the hell* is base-generated in Spec-AttP. Vu and Lohiniva (2020)’s derivation for (29) is shown in (30). At the start of the derivation, *the-hell* is base-generated high in Spec-AttP (30a). The nearest *wh*-phrase to *the-hell*, which is always the higher *wh*-phrase, moves to Spec-AttP to check the [+wh] feature (30b) before the full *wh-the-hell* phrase moves to Spec-CP (30c). In-situ *wh-the-hell* thus cannot be derived as *the-hell* would never modify the lower *wh*-phrase.

(29) Who the hell loves who?

¹⁴For completeness, we note that Den Dikken and Giannakidou (2002)’s account would predict that full and partial *wh*-movement questions should be able to host *the-hell*, which is consistent with what we find in §2. In both cases, as no *wh*-phrase intervenes between Q and *wh-the-hell*, Q is able to license *the-hell* and there are no intervention effects.

(30) Derivation for (29):

a. *the-hell* is base-generated in AttP:

[_{AttP} the hell Att_[+wh] [_{vP} who_[+wh] loves who]]

b. [+wh] feature checking:

[AttP who_i_[+wh] the hell Att_[+wh] [vP t_i loves who]]

A horizontal line connects the [+wh] feature on 'who' to the [+wh] feature on 'Att'.

c. Movement of *wh-the-hell* to Spec-CP:

[CP [who_i_[+wh] the hell]_j [AttP t_j Att_[+wh] [vP t_i loves who]]]

A horizontal line connects the Spec position of the CP to the position of 'the hell' in the AttP.

This analysis makes the correct predictions for *wh-the-hell* questions with full movement and *wh*-in-situ in Singlish but not for those with partial movement.

Wh-the-hell questions with full movement like (3a) follow a similar derivation to (30).

(3a) What the hell you think Natalie is baking at 3am ah?

‘What the hell do you think Natalie is baking at 3am?’

Vu and Lohiniva (2020)’s analysis is also able to rule out (3c). As the *wh*-phrase stays in situ, it cannot check the [+wh] feature in Spec-AttP and the derivation does not converge. In addition, it is not possible for *the-hell* in AttP (in IP) to modify the in-situ object *wh*-phrase in vP (31).

(3c) * You think Natalie is baking what the hell at 3am ah?

Intended: ‘What the hell do you think Natalie is baking at 3am?’

(31) * [AttP **the hell** Att_[+wh] [vP Natalie is baking **what**_[+wh] ...]]

However, Vu and Lohiniva (2020)’s analysis incorrectly rules out *wh-the-hell* in partial *wh*-movement questions in (3b), which our experiment in §2 finds to be acceptable.

(3b) You think what the hell Natalie is baking at 3am ah?

‘What the hell do you think Natalie is baking at 3am?’

In questions with partial wh-movement, the wh-element is moved to the embedded Spec-CP, and *the hell* is base-generated in the matrix Spec-AttP shown in (32). There is no way to generate the correct word order where the wh-element and *the hell* are both in the embedded Spec-CP.¹⁵

- (32) [IP you [AttP **the hell** Att_[+wh] [VP think [CP **what**_[+wh] Natalie is baking ...]]]]

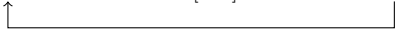
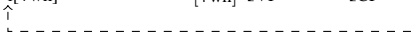
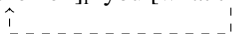
4 Conclusion

This paper establishes an empirical generalization for Singlish *wh-the-hell* questions through an acceptability judgment experiment: *the-hell* may modify wh-phrases that undergo full/partial movement, but cannot modify wh-in-situ in single wh-questions. To account for this, we assume that *the-hell* has a licensing condition which requires it to be in matrix Spec-CP to check its POV feature and ascribe its negative attitude to the speaker (Chou, 2012). *The-hell* is dependent on the movement of the wh-phrase to

¹⁵This analysis potentially incorrectly predicts the sentences in (i) to be acceptable in Singlish.

- (i) a. * You what the hell think Natatie is baking ah?
b. * You the hell think what Natatie is baking ah?

In (ia), *what* overtly moves from the embedded Spec-CP to Spec-AttP to check the [+wh] feature shown in (ii). In (ib), *what* covertly moves to Spec-AttP to check the [+wh] feature shown in (iii). In both sentences, *what-the-hell* then moves covertly from Spec-AttP to the matrix Spec-CP shown in (iv). The [+wh] feature on Att is satisfied and both sentences can be interpreted as questions (due to the covert movement to Spec-CP). Although the derivations can be generated under this analysis, the sentences are unacceptable.

- (ii) Derivation for (ia): overt movement of *what* to Spec-AttP
 [IP you [AttP what_[+wh] the hell Att_[+wh] [VP think [CP *t_i* Natalie is baking ...]]]]

 (iii) Derivation for (ib): covert movement of *what* to Spec-AttP
 [IP you [AttP what_[+wh] the hell Att_[+wh] [VP think [CP *what*_{*i*} Natalie is baking ...]]]]

 (iv) Derivation for (ia-b): covert movement of *what-the-hell* to Spec-CP
 [CP [what the hell]_{*i*} you [what the hell]_{*i*} Att think [CP Natalie is baking ...]]


reach matrix Spec-CP since *the-hell* is a modifier that adjoins to the wh-head (Merchant, 2002). This licensing condition is satisfied in full/partial movement questions as *wh-the-hell* undergoes overt/covert movement to matrix Spec-CP. However, since wh-in-situ is interpreted by unselective binding, the wh-phrase is bound by the Q operator and does not move covertly to matrix Spec-CP. As a result, the POV feature on *the-hell* cannot be checked and *wh-the-hell* is not licensed.

We considered two alternative accounts and find that neither is able to explain the full range of Singlish data. Den Dikken and Giannakidou (2002)’s intervention effects analysis predicts that full and partial wh-movement are able to host *the-hell* in single wh-questions, but their account does not explain the unacceptability of in-situ *wh-the-hell* in single wh-questions, regardless of whether the subject or object is queried. Vu and Lohiniva (2020) claim that *the-hell* is base-generated in AttP, which has a [+wh] feature that must be checked by the wh-phrase in its Spec position. This analysis correctly predicts that full wh-movement can host *the-hell* and that in-situ *wh-the-hell* in object wh-questions is unacceptable. However, this analysis does not allow for *the-hell* in partial movement questions, contrary to the experimental findings in §2.

The predictions of these analyses in §3.4 and our negative attitude ascription account are summarized in table 5. Our negative attitude ascription analysis fares better than previous analyses in explaining the full set of Singlish data as it explains the availability of *wh-the-hell* questions formed by full and partial wh-movement, as well as the unavailability of in-situ *wh-the-hell* subject and object questions. Our account builds on Chou (2012)’s analysis for Mandarin *the-hell*, *daodi*. Hence, even

though *wh-the-hell* and Mandarin *daodi* differ in whether the *wh*-phrase and *the-hell* must form a unit or may be discontinuous, evidence from Singlish suggests that their analyses can be unified.

	Long	Partial	Situ
Den Dikken and Giannakidou (2002)	✓	✓	✗
Vu and Lohiniva (2020)	✓	✗	✓
Negative attitude ascription proposal	✓	✓	✓

Table 5: Summary of predictions by various accounts

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